

Relating swimming performance of lake sturgeon, *Acipenser fulvescens*, to fishway design

S. Peake, F.W.H. Beamish, R.S. McKinley, D.A. Scruton, and C. Katopodis

Abstract: Fishways have traditionally been designed to provide safe passage for jumping fish and only recently have nonjumping species been considered. Concern over dwindling populations of lake sturgeon, *Acipenser fulvescens*, has focused attention on fishway designs that accommodate its swimming abilities. The objective of this study was to derive a model that relates swimming endurance of lake sturgeon to length and flow characteristics of fishways. Endurance at sustained and prolonged swimming speeds (those maintainable for more than 20 s) increased with water temperature but was independent of temperature at higher burst speeds. Endurance increased with total length at all swimming velocities. Swimming performance of lake sturgeon, relative to body length, is inferior to that of most salmonids, particularly at burst speeds. Fishway designers need to consider swimming ability, space requirements, and behavior of lake sturgeon to ensure that they can ascend potential migratory obstacles safely.

Résumé : Traditionnellement, les échelles à poissons ont été conçues pour permettre le passage sans danger des poissons capables de sauter et ce n'est que récemment que des espèces qui ne sautent pas ont été prises en considération. Les préoccupations concernant le déclin des populations d'esturgeons de lac, *Acipenser fulvescens*, ont attiré l'attention sur des échelles à poissons conçues pour accommoder ses capacités de nage. L'objectif de la présente étude était d'élaborer un modèle qui relie l'endurance à la nage de l'esturgeon de lac à la longueur et à l'écoulement des échelles à poissons. L'endurance à la nage prolongée et soutenue (qui peut être maintenue pendant plus de 20 s) a augmenté avec la température de l'eau, mais elle était indépendante de la température aux vitesses de pointe. L'endurance a augmenté avec la longueur totale à toutes les vitesses de nage. La performance de nage de l'esturgeon de lac, par rapport à sa longueur corporelle, est inférieure à celle de la plupart des salmonidés, particulièrement aux vitesses de pointe. Les concepteurs d'échelles à poissons doivent prendre en considération la capacité de nage, les besoins en espace et le comportement de l'esturgeon de lac pour s'assurer que ce dernier pourra franchir sans danger les obstacles potentiels à sa migration.

[Traduit par la Rédaction]

Introduction

Since the mid-1800's, populations of lake sturgeon, *Acipenser fulvescens*, have decreased dramatically due to commercial exploitation, loss of habitat, and an increasing sport fishery (Rochard et al. 1990). Slow growth, late sexual maturity, and infrequency of spawning all contribute to their vulnerability to exploitation (Brousseau 1987). In large rivers, hydroelectric installations can impede migrations and cause instability in flow rates, decreasing the quality of spawning and feeding habitat (Harkness and Dymond 1961; Auer 1996). Although fishways have been constructed on some rivers to allow safe passage of migratory fish, designs have been based primarily on the swimming capabilities of salmonids (Collins et al. 1962,

1963; Weaver 1963). Only recently have fishways been designed to accommodate other species (e.g., Katopodis et al. 1991).

Swimming ability among fish can vary with differences in anatomy and metabolism. Fishways designed to allow passage of one species can select against others of lesser ability (Schwalme et al. 1985). One difference that exists between sturgeon and salmonids is tail morphology. Salmonid tails are broad and homocercal whereas the tail of the lake sturgeon is heterocercal. The small lower lobe gives the sturgeon tail less depth, making its contribution to total thrust less than that of salmonids (Webb 1986). Sturgeon also produce more drag per unit of surface area than trout, presumably due to the presence of their bony plates or scutes (Webb 1986). Because drag increases with the square of velocity, high swimming speeds may impose a large energetic cost. Because routine metabolism of sturgeon has been found to be significantly lower than that of most teleost fishes (Singer et al. 1990), it may be that insufficient energy can be produced to attain the high speeds necessary to pass fishways designed for salmonid species.

Swimming activities of fish in general have been described by three categories: sustained, prolonged, and burst. Sustained occurs at relatively low velocities and represents speeds that can be maintained for a period greater than 200 min, making use of energy derived exclusively from aerobic processes (Beamish 1978). The velocity immediately below that that causes endurance to fall under 200 min is referred to as maximum

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sustained speed. Prolonged swimming covers a spectrum of speeds between sustained and burst. Burst represents high velocities that can be maintained for less than 20 s, using energy entirely generated by anaerobic processes (Beamish 1978). Sustained, prolonged, and burst speeds can be identified by measuring endurance over a range of swimming velocities. Endurance is defined as the amount of time a fish can swim at a particular speed.

Ideally, fishways should provide upstream passage for all migratory species inhabiting the river on which they are located. Water velocities within fishways should be less than the maximum attainable speed of all sizes and species that they are designed to pass, and the distance a fish must swim in the fishway should not exceed its endurance. The present study investigated swimming endurance of lake sturgeon in relation to body size and ambient temperature. Sustained, prolonged, and burst performance was compared with that of salmonids, and recommendations for future fishway designs capable of accommodating both species are offered.

Materials and methods

Lake sturgeon swimming performance was determined for individuals 12.0–132.0 cm in total length. A total of 58 sturgeon, 12–55 cm, were obtained as progeny of wild stock from the Moose River basin in northern Ontario. Five large fish, 106–132 cm, were captured in trap nets from southeastern Lake Huron. All fish were held in the laboratory in tanks supplied with running nonchlorinated well water. Sturgeon were subjected to a 16 h light : 8 h dark photoperiod. Small sturgeon (12–15 cm) were fed live earthworms to satiation three times per week, intermediate fish (23–55 cm) received daily feedings of dry pellet formula, and large fish (106–132 cm) were fed live crayfish (*Orconectes* sp.) to satiation three times per week.

Prior to experiments, water temperature was adjusted to 7.0, 14.0, or 21.0 ± 0.5°C by an increase of 1.0° or a decrease of 0.5° day⁻¹. Before fish were used in experiments, they were held at the selected temperature for at least 7 days and were not fed for 36 h immediately prior to testing to ensure that they were in a postabsorptive state. Swimming performance of small sturgeon was measured in 3.2-L swimming flumes (Waiwood and Beamish 1978). Intermediate fish were tested in a 200-L swimming flume (Farmer and Beamish 1969) modified by the addition of a vertical chimney at the downstream end of the swim chamber through which fish were introduced and removed. Water velocity within the flumes was measured using a recently calibrated current meter and controlled by rheostats. Water flow within the 3.2- and 200-L swim flumes was essentially rectilinear and even across the profile. Water from aerated and thermally controlled reservoirs was added at a rate of approximately 120 and 600 mL min⁻¹ for the 3.2- and 200-L flumes, respectively, to assure oxygen content in excess of air saturation and a constant temperature.

Performance of large sturgeon was measured in a black PVC pipe approximately 2.5 m long with a diameter of 56 cm. The tube was equipped with a fixed plastic retaining screen at the upstream end and a removable rubber mesh screen at the downstream end. Fish were introduced and removed at the downstream end of the tube. Two windows (15 × 25 cm) were cut in the apparatus about 0.5 m from each end to facilitate viewing of fish. Windows were covered from the inside with heavy-gauge clear vinyl to minimize internal turbulence. Strips of white tape were applied to the inside of the tube, in the window regions, to allow fish to be seen against the dark interior. After assembly, the entire apparatus was submerged and secured within a large aquatic flume (3 × 5 m) located at the University of Guelph, Guelph, Ont. Flow was found to be essentially rectilinear and flow rates were calibrated. Maximum cross-sectional areas of all lake sturgeon used in this study were less than 10% of that of their respective

swimming flumes, eliminating the need to adjust for blocking effect (Smit et al. 1971).

Performance was evaluated by measuring swimming endurance. Fish were transferred from a holding tank to a flume and allowed to swim at approximately 0.45 body lengths (bl)·s⁻¹ for 1 h, after which velocity was abruptly increased to the desired test velocity. Endurance was measured at velocities of 20, 40, 45, and 50 cm·s⁻¹ for small fish, ranging from 30 to 90 cm·s⁻¹ for intermediate sturgeon, and 90, 120, 150, and 180 cm·s⁻¹ for large fish. Small, intermediate, and large sturgeon were unable to swim at speeds greater than 50, 90, and 180 cm·s⁻¹, respectively. Small fish were tested at 7.0 and 14.0°C, intermediate sturgeon at 7.0, 14.0, and 21.0°C, and large fish at 14.0 ± 0.5°C. Velocities that resulted in endurance responses greater than 200 min were considered to represent sustained swimming. To ensure that sturgeon could maintain these speeds indefinitely, fish that swam in excess of 200 min were allowed to continue for an additional 280 min after which the experiment was terminated. Because these fish did not fatigue, but were interrupted, these data were not used in the development of the endurance model. Swimming endurance was estimated using multiple regression (Draper and Smith 1966) as follows:

$$\log(E) = a_0 + a_1L + a_2T + a_3V + a_4LT + a_5TV + a_6LV + a_7LTV + e$$

where E is endurance (minutes), L is total length (centimetres), T is water temperature (degrees Celsius), V is swimming velocity (centimetres per second), and e is a normally distributed error term with mean 0 and variance σ^2 . Values of P less than 0.05 indicated statistical significance.

Accurate measurements of swimming endurance depend on consistent recognition of fish fatigue. Electric shocking was not used in this study. Instead, fish were occasionally stimulated to maintain swimming by various methods, depending on the flume in which they were tested. Small fish were completely enclosed in the flume and were stimulated by short and sharp fluctuations in velocity followed by an immediate return to the test velocity. Intermediate fish, in the 200-L flume, were stimulated by gentle prodding with a plastic rod and by velocity fluctuations as described for small fish. Large sturgeon were also gently prodded to keep them from resting against the downstream retaining screen. In all cases, fish were considered fatigued when they became impinged on the downstream screen and failed to resume swimming despite repeated attempts to stimulate them. In most cases, onset of fatigue was obvious. After an experiment, fish were measured, weighed, and returned to their holding tanks. All fish were allowed to recover for at least 48 h prior to being used in another swim test at a different velocity.

Results and discussion

Swimming performance

Swimming endurance of lake sturgeon, in general, was found to increase with fish length at all swimming speeds and temperatures tested. Large fish could swim for longer and attain higher speeds than smaller fish. Swimming velocities that sturgeon could maintain for more than 200 min were considered to represent sustained speeds. The highest velocity that did not eventually result in fatigue represented maximum sustained speed for that fish. This speed can only be approximated experimentally, but was also found to increase with fish length.

The relationship between endurance, water temperature, fish length, and swimming speed is described by the regression

$$(1) \quad \log(E) = 1.40 + (2.26 \times 10^{-2} \times L) + (5.47 \times 10^{-2} \times T) - (4.55 \times 10^{-2} \times V) - (5.36 \times 10^{-4} \times T \times V) + (1.85 \times 10^{-4} \times L \times V).$$

The model has a critical F -value of 274.6 ($P < 0.05$), an adjusted r^2 of 0.854, and 233 degrees of freedom. Interactions between temperature and length and between temperature, velocity, and length were not significant.

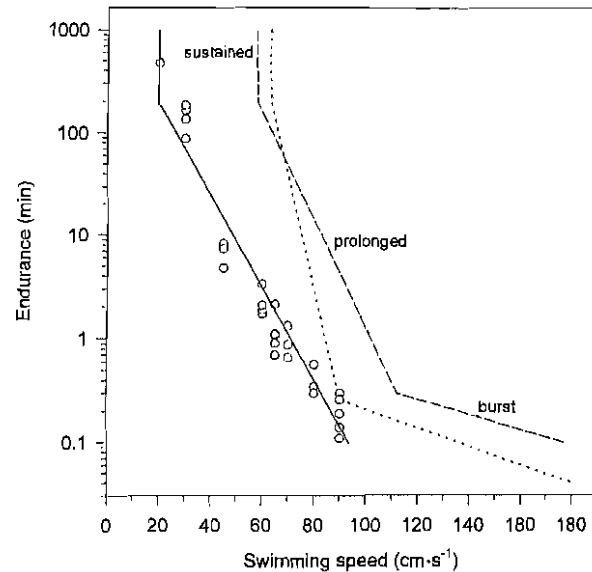
Fatigue curves are important tools in studies of fish swimming performance and, for lake sturgeon, can be generated by plotting endurance (calculated from equation 1) against a range of swimming speeds on a semilog graph. Fatigue curves, for salmonids, are characteristically resolved into three straight sections connected by two changes in slope (Fig. 1). The first section is vertical and is located at maximum sustained speed. Swimming speeds to the left of this section are considered sustained. The change in slope that occurs at maximum sustained speed marks the beginning of the second section which represents prolonged swimming. The second inflection typically occurs at a speed that results in fatigue after 20 s. This slope change marks the beginning of the third section representing burst swimming speeds.

Fatigue curves generated for lake sturgeon take the form of a single straight line with no change in slope between prolonged and burst swimming (Fig. 1). The line represents endurance for fish of a prescribed total length (12.0–132.0 cm), at a given temperature (7.0–21.0°C), at speeds greater than maximum sustained. It is not valid for endurance responses greater than 200 min. A vertical line must be added, at maximum sustained speed, to reflect the demonstrated ability of lake sturgeon to swim indefinitely at sustained velocities. This model is also not valid for determining endurance for fish greater than 55 cm long at temperatures other than 14°C, or for fish less than 23 cm long at temperatures greater than 14°C. A temperature of 14°C, for large sturgeon, should be appropriate for application to migrating fish which generally spawn when water temperatures are in the 11–16°C range (Table 1). Nevertheless, peak spawning has been observed at temperatures as low as 8.5°C (Kempinger 1988) and, in these cases, equation 1 may overestimate actual swimming capability.

Fatigue curves indicate that large sturgeon can swim longer and at higher speeds than smaller fish (Fig. 2). Sturgeon, 120 cm in length, at 14°C, can swim for 127.5 min at 90 $\text{cm}\cdot\text{s}^{-1}$, whereas 45-cm fish can maintain this speed for just 8.7 s. Maximum sustained speed also increases with length, from 4.0 $\text{cm}\cdot\text{s}^{-1}$ for 15-cm sturgeon to 83.7 $\text{cm}\cdot\text{s}^{-1}$ for 120-cm fish at 14°C. Endurance declines rapidly, for fish of all sizes, at velocities greater than their maximum sustained speed. A sturgeon 45 cm in length, at 14°C, can swim for 194.6 min at 20 $\text{cm}\cdot\text{s}^{-1}$, but will fatigue after 8.7 s at 90 $\text{cm}\cdot\text{s}^{-1}$.

Water temperature was also found to affect lake sturgeon swimming ability. Maximum sustained speed for a 45-cm fish increases with temperature, from 12.0 $\text{cm}\cdot\text{s}^{-1}$ at 7°C to 26.0 $\text{cm}\cdot\text{s}^{-1}$ at 21°C. This allows swimming categories for the same fish to change with a fluctuation in water temperature. The 45-cm sturgeon, swimming at 25.0 $\text{cm}\cdot\text{s}^{-1}$ and 7°C, fatigues after 59.8 min, indicating prolonged swimming, but at 21°C, the same fish can maintain 25.0 $\text{cm}\cdot\text{s}^{-1}$ for longer than 200 min, indicating sustained swimming. Endurance is also enhanced by temperature at prolonged speeds. Endurance of a 45-cm fish, swimming at 30 $\text{cm}\cdot\text{s}^{-1}$, increases with temperature, from 37.3 to 129.6 min at 7 and 21°C, respectively. The magnitude of the temperature effect is most pronounced at speeds just above maximum sustained, and decreases as velocity increases, eventually becoming inconsequential at high

Fig. 1. Comparison of fatigue curves for lake sturgeon (solid line), Arctic char, *Salvelinus alpinus* (dashed line), and rainbow trout, *Oncorhynchus mykiss* (dotted line) showing sections for sustained, prolonged, and burst swimming. Circles represent data for medium-sized sturgeon at 14°C. The fatigue curve for sturgeon was determined using a length value of 45 cm, a temperature of 14°C, and a range of swimming speeds (20–95 $\text{cm}\cdot\text{s}^{-1}$) substituted into equation 1. Positions of fatigue curves for rainbow trout (24 cm) and Arctic char (34 cm) were determined by Bainbridge (1960, 1962) and Beamish (1980), respectively.



speeds. The 45-cm sturgeon swimming at 90 $\text{cm}\cdot\text{s}^{-1}$ will fatigue after 7.8 s at 7°C and after 9.7 s at 21°C.

In general, lake sturgeon do not become sexually mature until they reach lengths of 100–150 cm (Table 1). From equation 1, a 130-cm fish can maintain position indefinitely in flows up to 96.8 $\text{cm}\cdot\text{s}^{-1}$, and they can swim for shorter periods at speeds up to 180 $\text{cm}\cdot\text{s}^{-1}$. These findings agree well with published reports concerning flow rates in areas where lake sturgeon spawn (Table 1).

Comparisons with salmonids

Patterns in lake sturgeon fatigue curves resemble those found for salmonids with some noteworthy exceptions. Curves representing sturgeon endurance do not show the second inflection corresponding to burst swimming. Endurance, at high speeds, simply continues to decline at the same rate as for prolonged speeds (Fig. 1). Consequently, it appears that lake sturgeon are not capable of attaining the high burst speeds that salmonids can typically achieve. This, however, is not to say that lake sturgeon do not possess burst swimming capability. Many species, whose performance has been shown to be affected by temperature, show the effect most profoundly at sustained and prolonged speeds, but not at burst (Brett 1964; Beamish 1978). Lake sturgeon endurance is similarly influenced by temperature, showing enhanced sustained and prolonged performance at higher temperatures, but an independence from temperature at speeds that can be maintained for less than 20 s. It is therefore probable that, at a

41

3.2 ft/s

5.9 ft/s

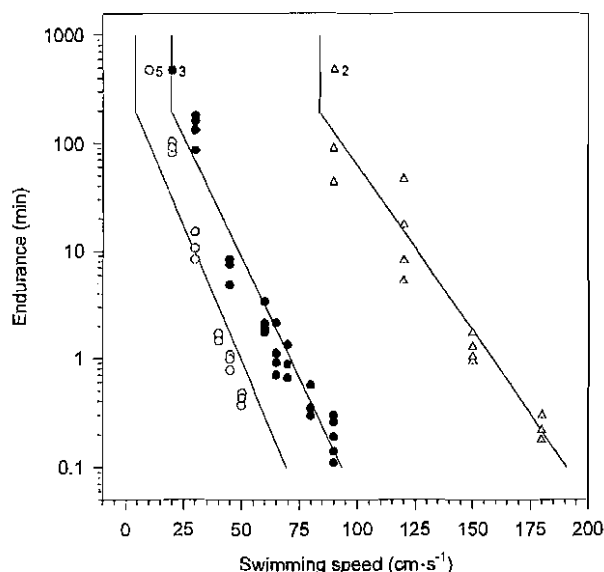
short periods

Table 1. Published values of flow speed, water temperature, and fish length at the time of sturgeon spawning (a dash indicates that the corresponding information was not provided).

Location	Flow speed (m·s ⁻¹)	Water temperature (°C)	Fish length (cm)	Source
Wolf River, Wisconsin, U.S.A.	>0.10	8.3–16.1	—	Kempinger 1988
Des Prairies River, Quebec, Canada	0.02–1.09	11.6–15.4	—	LaHaye et al. 1992
L'Assomption River, Quebec, Canada	0.4–1.39 (0.61–0.84) ^a	11.0–21.5	—	LaHaye et al. 1992
Southern Quebec, Canada	—	—	116.8	Roussow 1957
Sturgeon River, Michigan, U.S.A.	—	10–15	153.9	Auer 1996

^aFlows containing the highest egg concentrations.

Fig. 2. Relationship between length, endurance, and swimming speed of lake sturgeon. Open circles, solid circles, and open triangles represent data from small, intermediate, and large fish, respectively. Positions of fatigue curves (from left to right) were determined using length values of 15, 45, and 120 cm, swimming speeds of 4–190 cm·s⁻¹, and a temperature of 14°C substituted into equation 1. The number against each line (top) indicates the number of fish tested.



physiological level, these velocities represent burst swimming for lake sturgeon.

The poor swimming performance of sturgeon, relative to that of salmonids, is not confined to burst speeds but is evident in all categories of swimming. Studies of sturgeon physiology have found that metabolism is intermediate between the more primitive elasmobranch and the more advanced teleost fishes (Singer et al. 1990). A lower overall metabolism could partially account for the depressed swimming ability of lake sturgeon. Disproportionately poor performance at burst swimming speeds may also indicate that anaerobic processes used by sturgeon are less efficient at providing usable energy than those in salmonids. Further, the apparent absence of a change in slope, at burst speeds, may indicate a more gradual shift to exclusive anaerobic metabolism than that demonstrated by salmonids.

Morphology, and specifically that of the tail, can also significantly influence swimming capabilities among fish species. The salmonid tail is symmetrical, with upper and lower lobes producing similar thrust; however, the lower lobe of the sturgeon tail is smaller than the upper and consequently

generates 66% less thrust than the lower tail lobe of similarly sized trout (Webb 1986). The net effect is that the sturgeon tail, as a whole, generates 18% less thrust than that of trout over sustained and prolonged speeds (Webb 1986). The consequences of lower thrust are further complicated by morphological differences in body form between sturgeon and salmonids. Salmonids have smooth, streamlined bodies whereas sturgeon have a rough surface and a less fusiform shape. Studies have found that the drag generated by a sturgeon is approximately 3.5 times that of a trout of equal surface area (Webb 1986). This means that a sturgeon must generate more thrust, with its less efficient tail, than a trout and will require more energy to maintain the same swimming speed. Anaerobic metabolism, at burst speeds, provides a limited amount of energy that the sturgeon will likely use up faster than the trout, resulting in reduced burst performance.

Applications to fishways and culverts

Traditionally, fishways have been installed to provide safe passage for economically valuable salmonids, with little regard for other migratory species such as the lake sturgeon. If fishways are to be successful, fish must be able to ascend them quickly; therefore, prolonged and burst swimming capabilities are of primary importance in establishing suitable water velocities within the structures. Unfortunately, lake sturgeon swimming performance differs most from that of salmonids at burst speeds. For both species to use a fishway, water velocities must be maintained within the swimming performance characteristics of sturgeon, it being the weaker swimmer.

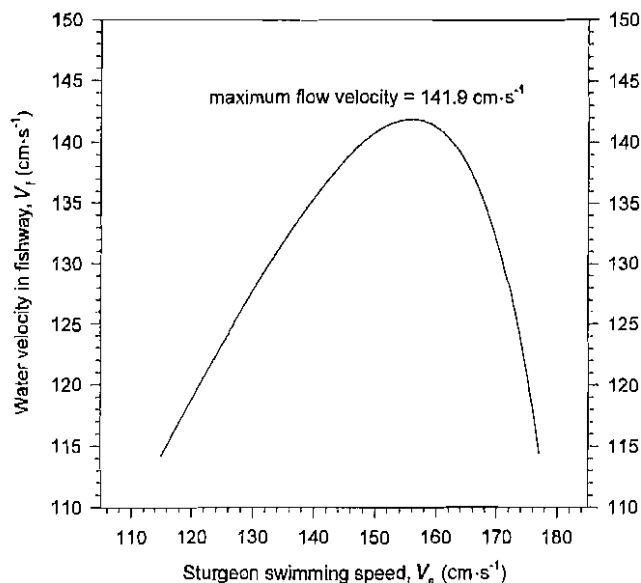
It must also be stressed that most studies, including this one, determine swimming performance using flumes where swimming speed is equal to water velocity. Fish ascending a fishway or culvert must attain positive ground velocity, which requires them to swim faster than the water velocity in the fishway. A formula can be used to determine water velocities that will allow sturgeon to ascend fishways, given the length of the fishway and the minimum size of fish to be passed:

$$(2) \quad V_f = V_s - (d \times E_v^{-1})$$

where V_f is the water velocity within the fishway (centimetres per second), V_s is the swimming speed of the minimum-sized sturgeon (centimetres per second), d is the length of the fishway (centimetres), and E_v is the endurance of the sturgeon swimming at V_s (seconds). E_v can be determined using equation 1.

The maximum fishway water velocity that is suitable for sturgeon passage can be determined using the second derivative of equation 2. This velocity can also be estimated by solving equation 2 for various values of V_s and plotting V_s on the x-axis and V_f on the y-axis. An example of this graphical

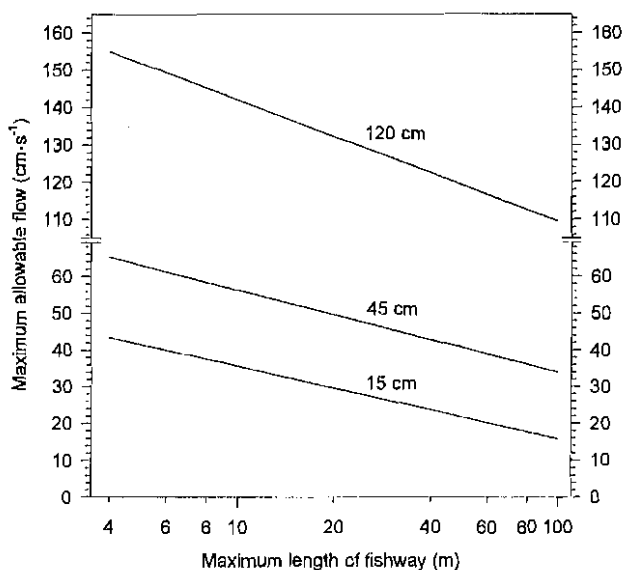
Fig. 3. Curve describing the swimming speeds required for a 120-cm lake sturgeon to successfully pass a 10-m fishway at 14°C. This curve was developed by substituting the appropriate values into equations 1 and 2. Under these circumstances, a 120-cm fish would not be expected to pass the fishway if flows within it exceeded 141.9 cm·s⁻¹. Different values of fish length, water temperature, and fishway length will yield different maximal values.



method is given in Fig. 3. Using this approach, maximum suitable velocity will correspond to the highest value of V_f on the resulting curve (141.9 cm·s⁻¹ in Fig. 3). If water velocity were to be set at 141.9 cm·s⁻¹, the 120 cm sturgeon would have to maintain a speed of 156 cm·s⁻¹ to successfully ascend the 10-m fishway. If, however, the velocity were set lower than maximum, for example at 135 cm·s⁻¹, the fish would be able to swim at speeds ranging from 140 to 168 cm·s⁻¹, thus creating a greater margin for error. Curves generated using equation 2 will change with temperature and fish length. Therefore, it is important, in determining E_{VS} , to use temperatures and lengths that reflect those of migrating lake sturgeon (Table 1). Fishway length and maximum water velocity combinations that are suitable for passing small, intermediate, and large sturgeon at 14°C are given in Fig. 4.

Finally, it is important to note that mature sturgeon can be much larger than salmonids utilizing the same fishway. This means that swimming performance of large spawning sturgeon may be similar to that of smaller salmonids. Therefore, if emphasis is to be placed on passing mature sturgeon, the physical dimensions of the structure may be more important than water velocity. Fishway designs requiring fish to jump from step to step are not generally conducive to the passage of sturgeon. If, however, the goal in establishing the fishway is to provide safe passage for sturgeon, regardless of size, then slowing water velocity becomes important to successful passage. If sturgeon populations are to recover in the future, it is important that they have access to the type of habitat utilized prior to construction of roads, hydroelectric dams, and other river impediments. If fishways and culverts are to accomplish this, new designs must reflect the unique swimming capabilities of the lake sturgeon.

Fig. 4. Lines representing the maximum flows that would allow 15-, 45-, and 120-cm lake sturgeon to pass fishways 4–100 m in length at 14°C.



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